

Seok-Young Kim

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Research Interests

Augmented Reality, 3D Scene Reconstruction/Generation, 3D Scene Understanding, Spatial Supersensing, HCI

Education

Korea Advanced Institute of Science and Technology (KAIST) Ph.D. in Graduate School of Metaverse	Mar. 2026 - Present Advisor: Woontack Woo
Korea Advanced Institute of Science and Technology (KAIST) M.S. in Graduate School of Metaverse Thesis: Object-level Interactive 3D Scene Generation from Physical-world Images in Mixed Reality	Mar. 2024 - Feb. 2026 Advisor: Woontack Woo
Technical University of Munich (TUM) Visiting Scholar - Computer Vision Group at the CAMP Chair hosted by Dr. Benjamin Busam - Topic: 3D Scene Understanding & Generation from Scene Graph	Dec. 2024 - Feb. 2025
Chonnam National University (CNU) B.S. in IoT Artificial Intelligence (<i>Summa Cum Laude</i>)	Mar. 2018 - Feb. 2024

Research Experience

Ubiquitous Virtual Reality Laboratory - Undergraduate Intern	Jan. 2024 - Feb. 2024 Advisor: Woontack Woo
Autonomous Intelligence Mobility Lab, CNU - Undergraduate Intern	Jan. 2022 - Dec. 2023 Advisor: Chansoo Kim

Publications

International Conferences and Journals

- [1] **Seok-Young Kim**, Jaewook Lee, Seokhwan Yang, Juyoung Lee, Jinseok Hong, Yoonseok Shin, Minju Baeck, Jon E. Froehlich, Woontack Woo
RealityCraftAR: In-Situ Authoring of Real-World Objects in Augmented Reality
Under Review, 2027
- [2] **Seok-Young Kim**, Abdelrahman Elskhawy, Taewook Ha, Dooyoung Kim, Eunjae Shin, Benjamin Busam, Woontack Woo
DeWorldSG: Depth-Aware 3D Semantic Scene Graph Generation via World-Model Priors
European Conference on Computer Vision (ECCV), 2026
- [3] **Seok-Young Kim**, Dooyoung Kim, Taejun Son, Youngbin Kim, Woojin Cho, Woontack Woo
Mix3R: Zero-shot Interactive 3D Scene Reconstruction from a Single Image in Mixed Reality
IEEE Transactions on Visualization and Computer Graphics (TVCG), 2026
- [4] **Seok-Young Kim**, Dooyoung Kim, Woojin Cho, Hail Song, Suji Kang, Woontack Woo
SceneLinker: Compositional 3D Scene Generation via Semantic Scene Graph from RGB Sequences
IEEE Transactions on Visualization and Computer Graphics (TVCG), 2026
- [5] Suji Kang, **Seok-Young Kim**, Youngbin Kim, Taewook Ha, Dieter Schmalstieg, Shohei Mori, Woontack Woo
SnapPhysics: A Physics-Aware Scene Graph from a Single View for Interactive Mixed Reality Scenes
IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2026

International Demos, Workshops and Posters

- [6] Jinseok Hong, Minju Baeck, **Seok-Young Kim**, Yoonseok Shin, Woontack Woo
Collaborative Scene Mood Authoring with Voice-driven Multimodal Feedback Design in Virtual Reality
ACM SIGGRAPH Asia 2025, XR Demo
- [7] **Seok-Young Kim**, Dooyoung Kim, Taejun Son, Woontack Woo
RealityCrafter: User-guided Editable 3D Scene Generation from a Single Image in Mixed Reality
ACM Symposium on User Interface Software and Technology (UIST 2025 Adjunct)

Domestic Conferences and Journals

- [1] **Seok-Young Kim**, Jinseok Hong, Minju Baeck, Woontack Woo
Scene Graph Diffusion Transformer for Controllable 3D Virtual Scene Generation
Korea Computer Congress (KCC) Conference, 2025 🏆 **Best Paper Award**
- [2] Suji Kang, Seokhwan Yang, **Seok-Young Kim**, Woontack Woo
Speech-to-3D: Personalized 3D Scene Rendering based on User Speech Recognition
Korea Computer Congress (KCC) Conference, 2025
- [3] Seungwoon Shin, **Seok-Young Kim**, Woontack Woo
Scene Graph-based Interactive 3D Scene Reconstruction from RGB Sequences
Korea Software Congress (KSC) Conference, 2024
- [4] **Seok-Young Kim**, Chansoo Kim
Improved Depth Completion with a Two-branch Backbone based on CNN-ViT integration module
Transactions of the Korean Society of Automotive Engineers, KCI, 2024
- [5] **Seok-Young Kim**, Chansoo Kim
TB-CompletionFormer: Improved Depth Completion based on Two-branch Backbone
Korean Society of Automotive Engineers (KSAE) Annual Fall Conference, 2023
- [6] **Seok-Young Kim**, Yeonggyu Park, Taehyun Park, Yuri Seo, Seongjun Kim, Kichun Jo, Chansoo Kim
Towards precise Depth Completion guided by dense Pointcloud based on LiDAR Accumulation
Korean Society of Automotive Engineers (KSAE) Annual Spring Conference, 2023

Teaching Experience

URP490: Undergraduate Research Participation Program	School of Computing, KAIST
- Student: Seungwoon Shin	Fall 2024
- Material: One KSC'24 paper	

Projects

Development of Technology to Reenact Industrial Spatial Knowledge from Incomplete Data	
UVR Lab, KAIST	Apr. 2026 - Present
Collaborative Interfaces for AR Content Authoring Among SpaceTop Users	
SpaceTop Research Center, KAIST	Dec. 2024 - Present
DT-XR: Development of Dynamic Digital Twin for Realistic Untact XR Collaboration	
UVR Lab, KAIST	Mar. 2024 - Feb. 2026

Awards and Honors

Best Paper Award

Korea Computer Congress (KCC) Conference	2025
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Grand Prize, Artificial Intelligence System Competition

Chonnam National University	Nov. 2022
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Silver Prize, International Electric Vehicle Expo Autonomous Driving Competition

International e-Mobility Expo	Apr. 2022
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Scholarship for Academic Excellence

Chonnam National University	All semesters of 2021-2023
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